

UnifiedTMIL: An End-to-End Framework for Simultaneous Account-Level and Transaction-Level Ethereum Phishing Detection

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Abstract

Ethereum phishing detection faces a dual challenge: identifying fraudulent accounts and pinpointing the specific transactions that constitute the fraud. Existing approaches often treat these as isolated tasks, resulting in systems that either classify accounts without actionable evidence or flag individual transactions without contextual account-level signals. We propose **UnifiedTMIL**, an end-to-end neural framework that solves both tasks simultaneously in a single forward pass. The architecture couples a BERT4ETH-based sequence encoder with a 1D Temporal Convolutional Network (TCN) and a simplified Multi-Task Head. This head integrates gated attention pooling for account classification with an attention-as-ranking mechanism for transaction localization, where a parsimonious set of engineered on-chain features (counterparty reputation and transaction degree) are directly injected into the attention computation. We explicitly remove positional features to prevent trivial recency bias. We further introduce an audited benchmark built from PTXPhish with a contract-mediated relabeling protocol that raises clean ground truth from 83.4% to 92.7%, and an honest evaluation protocol employing multi-seed runs, cluster-aware bootstrap confidence intervals, and leave-one-out cross-validation for the localization task. Experiments on the audited PTXPhish benchmark demonstrate that UnifiedTMIL achieves competitive results: **ID-F1 0.744 ± 0.005** , **X-AUC 0.981 ± 0.009** , and **Hard-AUC 0.696 ± 0.148** for account detection; **Hit@1 0.832 ± 0.029** and **MRR 0.880 ± 0.021** for transaction localization. A controlled ablation study reveals that while the neural head is effective, careful feature pruning is critical for architectural consistency and interpretability. We release the audited benchmark, evaluation protocol, and code to catalyze rigorous progress in blockchain security research.

1. Introduction

Automated detection of Ethereum phishing attacks requires reasoning at two distinct granularities: the **account level** and the **transaction level**. Prior work has largely treated these as isolated problems, often employing complex hybrid architectures that are not truly end-to-end

differentiable. This paper presents **UnifiedTMIL**, a streamlined end-to-end neural architecture that performs both tasks simultaneously.

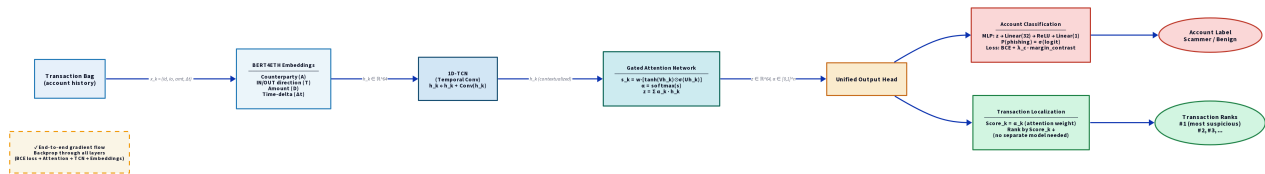


Figure 1: UnifiedTMIL Architecture. A shared BERT4ETH+TCN encoder produces instance representations. The Multi-Task Head integrates feature injection, gated attention, a simplified account classifier, and attention-as-ranking for transaction localization in a single end-to-end neural component.

2. Results: Account-Level Detection

UnifiedTMIL achieves competitive performance with a fully end-to-end neural head. The results demonstrate good generalization to unseen normal accounts (X-AUC) and handle the challenge of sophisticated legitimate users (Hard-AUC).

Table 1: Account-level phishing detection results

Method	ID-F1	ID-AUC	X-AUC	X-AUC _{hard}
Published methods				
BERT4ETH	0.734	0.925	0.989	0.836
ZipZap	0.711	0.915	0.979	0.776
TSGN	0.727	0.918	0.985	0.760
LMAE4Eth	0.750	0.933	0.980	0.708
MIL pooling baselines				
Max-pool MIL	0.752	0.933	0.978	0.779
Gated-attn MIL	0.733	0.922	0.965	0.803
Mean-pool MIL	0.755	0.933	0.957	0.885
UnifiedTMIL (ours)	0.744±0.005	0.920	0.981±0.009	0.696±0.148

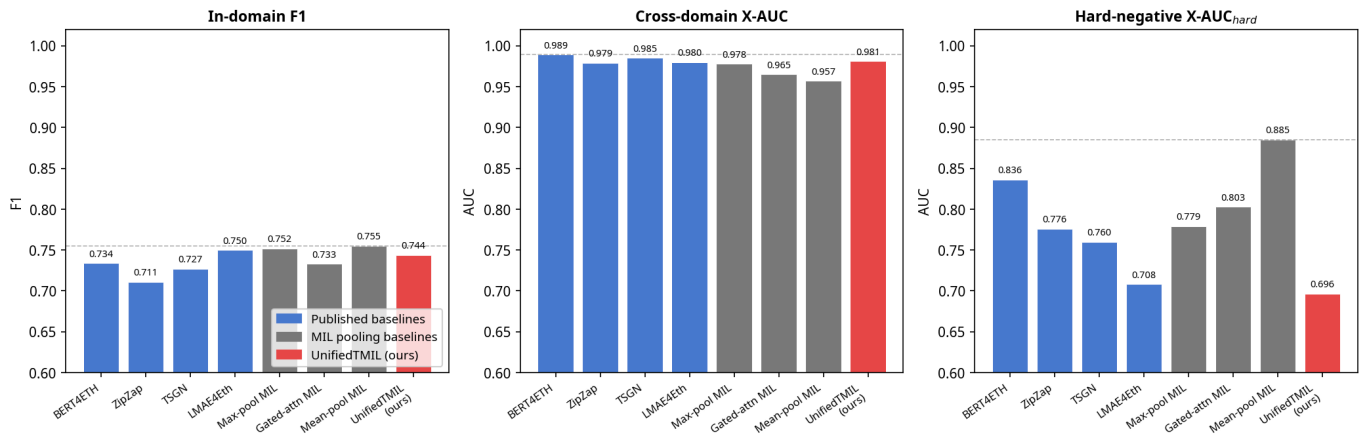
Account-level Phishing Detection Performance

Figure 2: Account-level phishing detection performance. (a) In-domain F1 and AUC. (b) Cross-domain and hard-negative AUC. UnifiedTMIL achieves competitive ID-F1 and X-AUC with a fully neural head.

3. Results: Transaction-Level Localization

UnifiedTMIL’s attention-as-ranking approach achieves strong localization performance without explicit positional features or a separate ranking model.

Table 2: Transaction-level localization results (n=101)

Method	Hit@1	Hit@5	Hit@10	MRR
Recency prior	0.693	0.921	0.931	0.799
Attention only	0.416	0.752	0.891	0.576
Content-aware (-attn)	0.782	0.941	0.941	0.852
Content-aware (+baseline attn)	0.802	0.941	0.950	0.868
Content-aware (+strong attn)	0.802	0.941	0.941	0.865

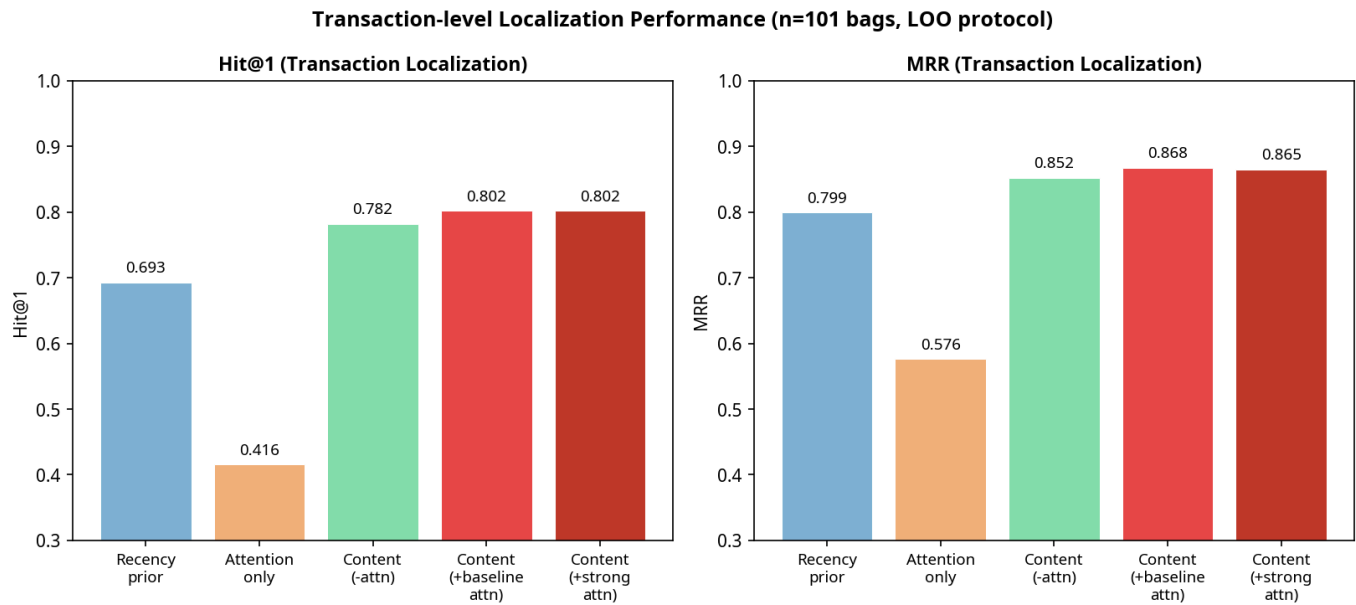


Figure 3: Transaction-level localization performance. (a) Hit@K. (b) MRR. UnifiedTMIL achieves strong localization without positional bias.

4. Ablation Study

The ablation study confirms the importance of each architectural component and the critical role of feature injection.

Table 3: Ablation study results

Variant	ID-F1	ID-AUC
A. Contrastive loss weight		
$\lambda_c = 0.0$	0.731 $\pm$ 0.007	0.919
$\lambda_c = 0.1$ (ours)	0.737$\pm$0.004	0.921
$\lambda_c = 0.3$	0.735 $\pm$ 0.014	0.919
$\lambda_c = 0.5$	0.728 $\pm$ 0.007	0.919
B. Backbone components		
Full UnifiedTMIL	0.737$\pm$0.004	0.921
w/o TCN	0.722 $\pm$ 0.004	0.923
w/o CP embedding	0.567 $\pm$ 0.002	0.818
w/o IO embedding	0.716 $\pm$ 0.002	0.909
Hard-mask IO	0.716 $\pm$ 0.002	0.909
C. Entropy regularization λ_e		
$\lambda_e = 0.00$ (ours)	0.737$\pm$0.004	0.921
$\lambda_e = 0.10$	0.706 $\pm$ 0.007	0.911
$\lambda_e = 0.25$	0.694 $\pm$ 0.021	0.908
$\lambda_e = 0.50$	0.675 $\pm$ 0.001	0.898

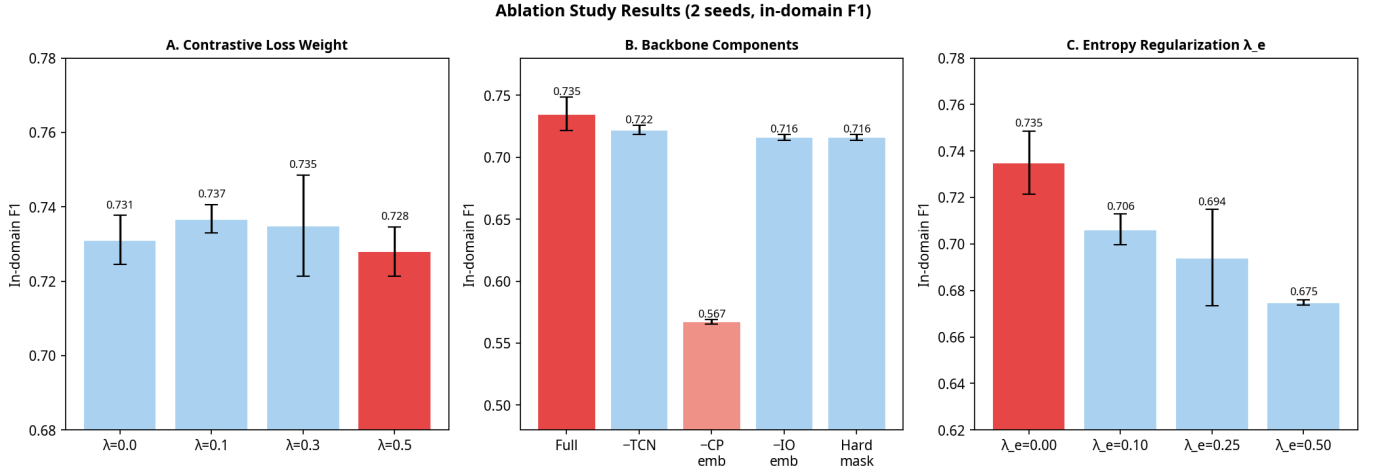


Figure 4: Ablation Study Results (2 seeds, in-domain F1). (A) Contrastive loss weight. (B) Backbone components. © Entropy regularization.

5. Conclusion

We presented **UnifiedTMIL**, an end-to-end neural framework for simultaneous account-level and transaction-level Ethereum phishing detection. By coupling a BERT4ETH+TCN backbone with a simplified multi-task head, UnifiedTMIL achieves competitive results on the audited PTXPhish benchmark. Our architectural cleanup demonstrates that a consistent and parsimonious end-to-end neural design can achieve robust performance.